A

22,393 pairs from 408 target genomes
(579 of the 727 anchor genes contributed at least one)

same genus
272 pairs · 1 genome
same family
7,104 pairs · 55 genomes
same order
6,472 pairs · 41 genomes
same class
8,546 pairs · 311 genomes
0
2000
4000
6000
8000
10000
12000
orthologue pairs

B

Where each rank sits on the Fig. 1a axis

same genus (median 94%)
same family (median 51%)
same order (median 46%)
same class (median 39%)
pairs within rank (%)
0
5
10
15
20
25
30
30
40
50
60
70
80
90
100
gene relatedness (identity at third codon positions, %)

C

The association survives holding ancestry fixed

orthologue
non-homologous gene, same genome
random downstream flank, same genome
dinucleotide shuffle of the orthologue upstream
Spearman ρ within rank (similarity vs gene relatedness)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.35
0.55
0.21
0.05
same genus
same family
same order
same class

D

The association does not depend on the statistic

series as in (C)
D2*
Pearson
Spearman ρ (all pairs)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
k=3
k=4
k=5
k=3
k=4
k=5